Another Life

The third way to change the laws of life is to engineer completely inorganic beings. The most obvious examples are computer programs that can undergo independent evolution.

Recent advances in machine learning already enable present-day computer programs to evolve by themselves. Though the program is initially coded by human engineers, it can subsequently acquire new information on its own, teach itself new skills and gain insights that go beyond those of its human creators. The computer program is therefore free to evolve in directions its makers could never have envisaged.

Such computer programs can learn to play chess, drive cars, diagnose diseases and invest money in the stock market. In all these fields they might increasingly outperform old-fashioned humans, but will have to compete against each other. They will thus confront new forms of evolutionary pressures. As a thousand computer programs invest money in the stock market, each adopting different strategies, many will go bust but some will become billionaires. In the process, they will evolve remarkable skills that humans can neither rival nor understand. Such a program could not explain its investment strategy to a Sapiens, for the same reason that a Sapiens could not explain Wall Street to a chimpanzee. Many of us might eventually work for such programs, which will decide not only where to invest money, but also whom to hire for a particular job, whom to give a mortgage to and whom to send to prison.

Are these living creatures? It depends on what you mean by ‘living creatures’. They have certainly been produced by a new evolutionary process, completely independent of the laws and limitations of organic evolution.

Imagine another possibility – suppose you could back up your brain to a portable hard drive and then run it on your laptop. Would your laptop be able to think and feel just like a Sapiens? If so, would it be you or someone else? What if computer programmers could create an entirely new but digital mind, composed of computer code, complete with a sense of self, consciousness and memory? If you ran the program on your computer, would it be a person? If you deleted it could you be charged with murder?